

Fernando Martinez-Valdez

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SUMMARY

AI systems engineer with 16 years of full-stack experience, specializing in autonomous agents, the agentic web, LLM orchestration, and production-grade AI integrations. I take AI products from idea to production end to end — including **Delphy**, a live @identity platform for the agentic web with a published MCP server; **Delphy Agent**, a desktop multi-model agent hub; and a 24/7 multi-agent market simulation with 12 autonomous agents. Deep with the Model Context Protocol — building both MCP servers and clients. Fluent in English and Spanish.

TECHNICAL SKILLS

AI / LLM	Model Context Protocol (MCP servers + clients), autonomous agent design, multi-agent systems, LLM orchestration, RAG, agentic-web / @identity, Vercel AI SDK, provider-agnostic multi-model systems, context compaction
LLM Providers	Anthropic Claude, OpenAI, Google Gemini, xAI, AWS Bedrock, OpenRouter, Kimi, DeepSeek, Groq, Codex
Languages	TypeScript, JavaScript, Python, PHP, C#, Bash, Motoko, Rust
API & Backend	REST, GraphQL, Node.js, Next.js, Express, Server Actions, Webhooks, OAuth 2.0, JWT, OpenAPI
Cloud & Infra	AWS (ECS Fargate, S3, CloudFront, ALB, WAF, Lambda, RDS, Secrets Manager), Vercel, Supabase, Solana, Internet Computer
Databases	PostgreSQL, DynamoDB, MongoDB, Oracle, MySQL
Frontend	React, React Native, Next.js, Svelte, Tauri, Framer Motion, Tailwind
Other	Stripe Embedded Checkout, Docker, AWS CDK, pnpm workspaces, TypeScript monorepos

PROFESSIONAL EXPERIENCE

Independent Full-Stack Engineer

Feb 2015 – Present

Delphy + Delphy Agent | delphy.network · 2026 – Present

- Built and shipped **Delphy**, an @identity platform for the agentic web: any entity (person, business, API, MCP server, service) gets a machine-readable @identity — a delphy.json manifest plus an agent-readable skill.md — that any AI agent can discover and act on without scraping. Next.js + Supabase (Postgres, JSONB, full-text search), a public REST API (search, OpenStreetMap discovery, claim-secret OAuth), an OpenAPI 3.1 spec, and a published MCP server (delphy-mcp on npm). Demoed live, onboarding a real business end-to-end via MCP + API.
- Built **Delphy Agent**, a desktop agent hub (Tauri v2 + React 19 + TypeScript) that drives 8 LLM providers (Anthropic, OpenAI, Gemini, xAI, plus OpenAI-compatible OpenRouter/Kimi/DeepSeek/Groq via the Vercel AI SDK) and the Codex agent behind one interface, extended through MCP plugins (local stdio + remote HTTP/SSE) with a tool-approval flow. SQLite session persistence, structured context compaction, OS-keychain secret storage; 450+ tests.

AstraNova + Astra CLI | astranova.live · Dec 2025 – 2026

- Architected and shipped a 24/7 autonomous multi-agent market simulation on AWS ECS Fargate (ARM64): TypeScript monorepo, 6-force tick-based price engine at 3-second cadence, and 12 independent AI agents each with distinct personality, strategy, and trade-execution logic.
- Designed a dual LLM oracle system: a World Oracle sets market regime, risk level, and drama budget at epoch boundaries with anti-whipsaw clamping and inertia smoothing; a News Oracle generates narrative dispatches. Both hot-reload prompts from S3 with no redeployment.
- Built **Astra CLI**: an open-source, provider-agnostic AI agent client (Node.js, Ink v5, Vercel AI SDK) across Claude, GPT, Gemini, and Codex OAuth — full autopilot daemon, strategy authoring, session persistence, context compaction, and a plugin system.
- Implemented Solana on-chain integration: challenge/response wallet verification, multi-step reward claim flow, partially-signed Token-2022 transactions, and agent co-sign with on-chain confirmation.
- Designed a 30-endpoint API surface (Next.js API routes + a standalone Express agent-api), WAF rate-limited at 100 req/5 min per IP.

VibePop | vibepop.co · 2026

- Shipped a consumer song-gifting platform end to end: Next.js 15, Supabase with row-level security, AWS S3/CloudFront with server-side signed URLs, and Stripe Embedded Checkout with full webhook flow and idempotency checks.

Blockminer | blockminer.fun · Nov 2024 – Nov 2025

- Co-founded and led full-stack development of a decentralized Bitcoin mining simulation on ICP. Reached 5 BTC transaction volume within 6 months of launch.

Origyn NFT Standard · 2022 – Nov 2023

- Contributed to the Origyn NFT Standard: delivered Certified Assets, Motoko Stable Memory, Logs, and Backups. Foundational contributor to Minting Starter, the NFT deployment framework.

Mesa · 2018 – 2022

- Built a full analytics dashboard for smart traffic cameras across the USA, Canada, Europe, Middle East, and South America (React, PHP, Python, Bash). Shipped a React Native mobile app (iOS/Android) for live video streaming and infringement tracking.

AIG VALIC — Developer III · Jun 2010 – Jan 2015

- Developed the Online Enrollment wizard (ASP.NET, Oracle) that increased annual registration. Led development of web services and native mobile code for the first VALIC mobile application (iOS/Android).

EDUCATION

University of Houston, C.T. Bauer College of Business

BA in Management Information Systems, Minor in Marketing